

HiHat Parameter Guide

Parameter reference — Thee Coast Machine

1. Internal Structure

White noise + metal oscillators → bandpass → drive. Includes the TR-808 set of six square waves at 215/304/364/467/542/681 Hz.

Element	Role	Sonic character
White noise	Broadband base material	Soft hiss
Metal oscillator (6 squares)	TR-808 metallic resonances	"shing", "tk"
Bandpass filter	Picks out a frequency band (FREQ + Q)	Tonal character
Drive (tanh)	Saturation for aggression	Edgy, cutting
Closed / Open	MIDI notes select decay time (mutually exclusive)	Short vs. long

2. Parameter Reference

FREQ (3000 — 14000 Hz)

Center frequency of the bandpass.

- **3000–5000 Hz:** darker, warmer
- **5000–8000 Hz:** mid-tilted, felt-like
- **8000–10000 Hz:** standard House / Techno
- **10000–14000 Hz:** shimmering, Modern / EDM

Tip: Keep 2–3 kHz separation from Snare / Clap.

Q (1 — 12)

Bandpass sharpness.

- **1–2:** broad
- **2–4:** natural hat
- **4–7:** pinpoint
- **7–12:** heavy resonance

Tip: High Q reduces level. Compensate by lowering FREQ.

METAL (0 — 1)

Mix amount of metal oscillator.

- **0.0:** pure noise

- **0.3–0.5**: subtle edge (808-style)
- **0.5–0.8**: clearly metallic
- **0.9–1.0**: mostly metal

Tip: Pairs well with Q for character variation.

DRIVE (0 — 1)

Tanh saturation after bandpass.

- **0.0**: clean
- **0.1–0.3**: subtle weight
- **0.4–0.7**: distinct distortion
- **0.8–1.0**: noisy, rough

Tip: Can stand in for drum-bus tape saturation.

CLOSED DECAY (0.01 — 0.4 sec)

Closed hat decay.

- **0.01–0.03**: very tight
- **0.03–0.06**: standard closed
- **0.06–0.15**: shuffle feel
- **0.15+**: practically open

OPEN DECAY (0.05 — 1.5 sec)

Open hat decay. Triggered by MIDI note 92 (G#5).

- **0.05–0.15**: short open
- **0.15–0.4**: standard open
- **0.4–0.8**: long, jazzy
- **0.8–1.5**: very long tail

Tip: Closed/Open are mutually exclusive (closed chokes the ringing open).

3. Character Presets

Goal	FREQ	Q	METAL	DRIVE	C-DEC
TR-808 style (warm)	8500	3.0	0.30	0.00	0.04
TR-909 style	9500	2.5	0.45	0.10	0.045
TR-606 style (Electro)	7500	4.5	0.75	0.05	0.035
Acid / Hardfloor	8500	3.5	0.40	0.25	0.045
Industrial	8000	6.0	0.60	0.70	0.035
Footwork / DnB	10500	2.5	0.30	0.00	0.025
Lo-Fi / Boom Bap	6500	2.5	0.10	0.00	0.055
Modern Tech House	10000	2.2	0.35	0.10	0.045
Ambient / Dub	6000	2.0	0.10	0.00	0.080

4. Tuning Order

#	Do this	Why
Step 1	Place FREQ away from other top-end	Sit above Snare or Clap?
Step 2	Set CLOSED DECAY	Tight or shuffly?
Step 3	Dial METAL	Soft, or mechanical?
Step 4	Sharpen with Q	Broad or pinpoint?
Step 5	DRIVE for aggression	More cut, more grit

5. Troubleshooting

Symptom	Cause & fix
Hat gets buried	Separate FREQ by 2–3 kHz, or use DRIVE.
Painfully shiny	Lower FREQ, Q, or HhVol.
Too mechanical	METAL below 0.3, lower Q.
No open hat	Send MIDI note 92 (G#5) on ch 10.
Closed and open ring together	Mutually exclusive — intended choke behaviour.

※ The hi-hat carries the groove. If the kick is "boom", the hi-hat writes the "tick-tack".